volts during normal usage.

Through resistors R27 and R28, the output voltage of IC4 is set to 5V. Capacitors C4 through C6 are used for stability of IC4. The 5V rail thus obtained is used to power the rest of the circuit.

LEDs and USB charging. Circuit connections for power LEDs and the USB charging socket are shown in Fig. 4.

Here two pairs of 3x1-watt power LEDs are used for lighting application. A 2-ohm, 1W current-limiting resistor is used to limit current in each pair of LED channels in series with on/off switches S2 and S3. These LED channels receive power directly from the non-current-limited 5V rail (from the power unit). TPS2051C, which is a 500mA current-limited load switch, is used to drive a 500mA-limited 5V rail to power/charge the USB application.

We have used a current-limited load switch to prevent the USB application from fast draining the clock battery. However, it is optional. Capacitor C7 is used for stability of the current limiter.

For USB charging applications, a female USB-A type jack is used. It receives power from the 500mA, 5V current-limited channel of the power unit.

Main controller clock circuit. Circuit of the microcontroller-based main clock circuit is shown in Fig. 5. Apart from general time display, it also enables alarm function and indication of date, month, year and temperature.

The circuit is built around ATmega8 (IC8) microcontroller. IC8 is used to update the display, sense user inputs and take decisions. ATmega8 has 8kB of in-system programmable Flash with read-while-write capabilities, 512 bytes of EEPROM, 1kB of SRAM, 23 general-purpose input/output (I/O) lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte-oriented two-wire serial interface, a 6-channel analogue-to-digital converter (ADC) with 10-bit accuracy, a programmable watchdog timer with internal oscillator, an SPI serial port and five software-selectable power-saving modes, which are sufficient enough for this application.

A separate in-circuit serial programming port (CON12) is available for programming the microcontroller. Time keeping is done by RTC chip BQ32000 (IC7), which provides an automatic backup supply using a 3V battery. IC BQ32000 has a programmable calibration adjustment from –63ppm to +126ppm, so that it can also be used with low-quality crystals. It includes automatic leap-year compensation as well. The RTC chip is connected to the microcontroller via the I2C bus. This project uses a 3V CR2032 lithium cell

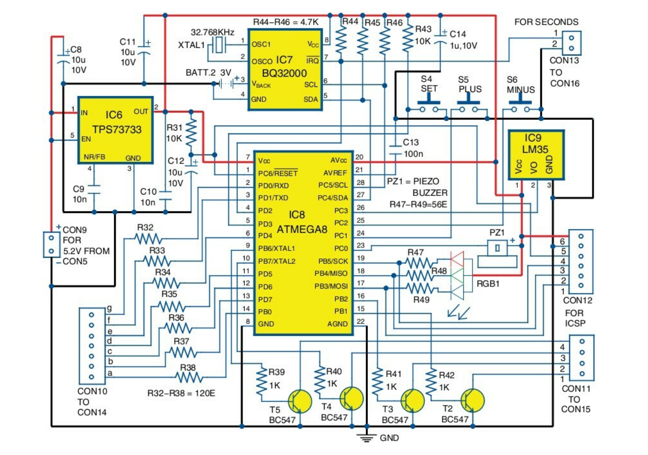

Fig. 5: Main microcontroller clock circuit